

Experiment 5: Vacuum Cleaner Problem

Aim

To implement and solve the Vacuum Cleaner Problem using Artificial Intelligence techniques.

Algorithm

1. Start
2. Initialize the problem state
3. Apply valid operations/search
4. Repeat until goal state is reached
5. Display solution
6. Stop

Flowchart

```
graph TD
    Start([Start]) --> Init[Initialize]
    Init --> Process[Process/Search]
    Process --> Goal{Goal Reached?}
    Goal -- Yes --> Print[Print]
    Goal -- No --> Repeat[Repeat]
    Repeat --> Process
    Print --> Stop([Stop])
```

Objective

- Understand state-space search.
- Implement AI search technique.
- Obtain the required goal state.

Program

```
rooms = {}
```

```
rooms["A"] = "Dirty"
```

```
rooms["B"] = "Dirty"
```

```
for room in rooms:
```

```
if rooms[room] == "Dirty":  
  
    print("Cleaning", room)  
  
    rooms[room] = "Clean"
```

```
print(rooms)
```

Output



The screenshot shows a code editor interface with a dark theme. On the left, there is a sidebar with icons for file explorer, search, and other tools. The main area displays a Python file named 'main.py' with the following code:

```
1 rooms = {}  
2  
3 rooms["A"] = "Dirty"  
4 rooms["B"] = "Dirty"  
5  
6 for room in rooms:  
7     if rooms[room] == "Dirty":  
8         print("Cleaning", room)  
9         rooms[room] = "Clean"  
10  
11 print(rooms)  
12
```

On the right side of the editor, there is an 'Output' panel. It shows the execution results of the code:

```
Cleaning A  
Cleaning B  
{'A': 'Clean', 'B': 'Clean'}  
  
=== Code Execution Successful ===
```

Result

The program was executed successfully.

Conclusion

The program demonstrated the AI concept successfully and produced the expected result.